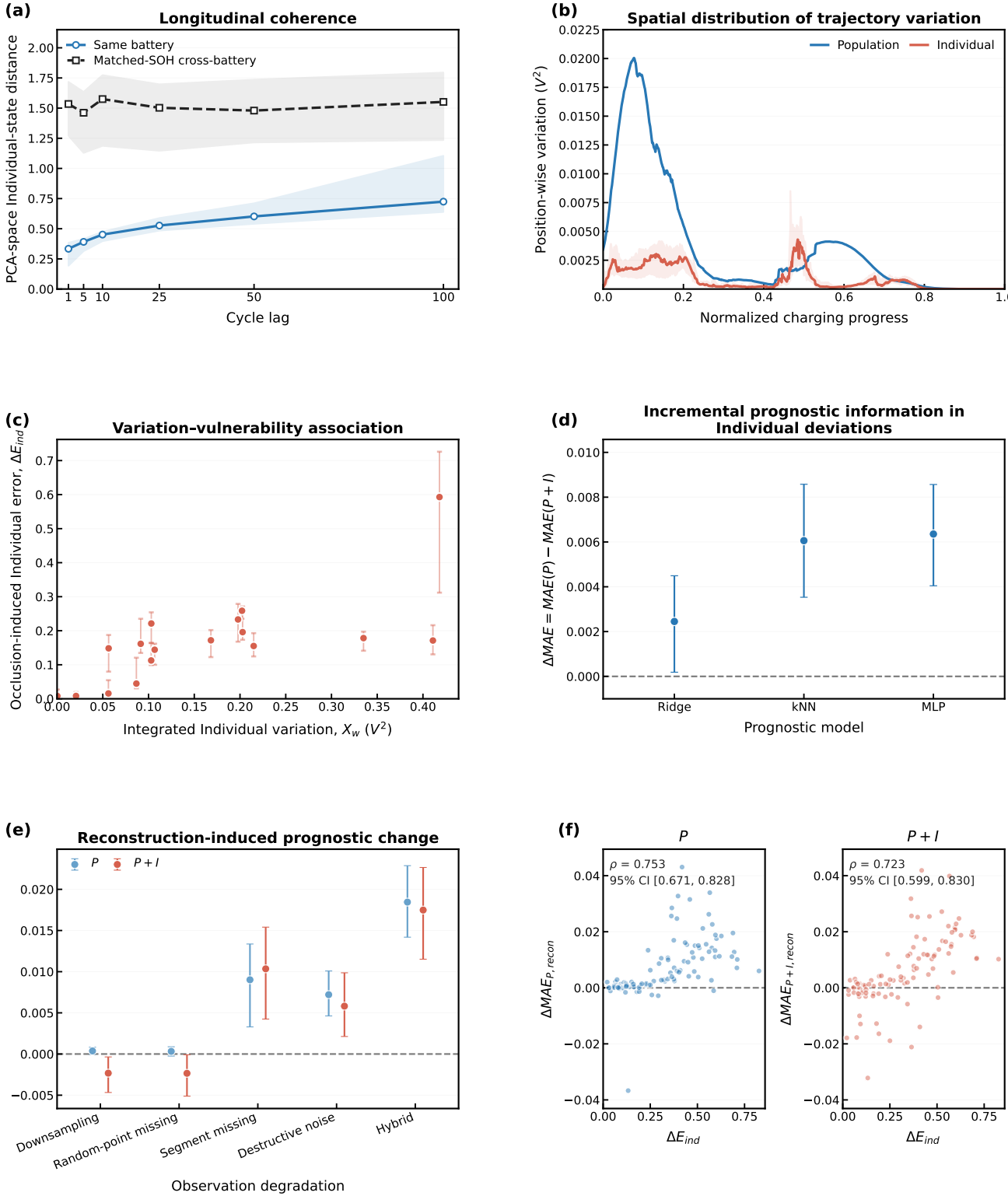

(a)
Longitudinal coherence
Same battery
Matched-SOH cross-battery
PCA-space Individual-state distance
Cycle lag
(b)
Spatial distribution of trajectory variation
Population
Individual
Position-wise variation (V^2)
Normalized charging progress
(c)
Variation–vulnerability association
Occlusion-induced Individual error, ΔE_{ind}
Integrated Individual variation, X_w (V^2)
(d)
Incremental prognostic information in Individual deviations
$\Delta MAE = MAE(P) - MAE(P+I)$
Ridge
kNN
MLP
Prognostic model
(e)
Reconstruction-induced prognostic change
P
$P+I$
ΔMAE_{recon} $= MAE_{recon} - MAE_{clean}$
Downsampling
Random-point missing
Segment missing
Destructive noise
Hybrid
Observation degradation
(f)
P
$P+I$
ρ = 0.753
95% CI [0.671, 0.828]
ρ = 0.723
95% CI [0.599, 0.830]
$\Delta MAE_{P,recon}$
$\Delta MAE_{P+I,recon}$
ΔE_{ind}